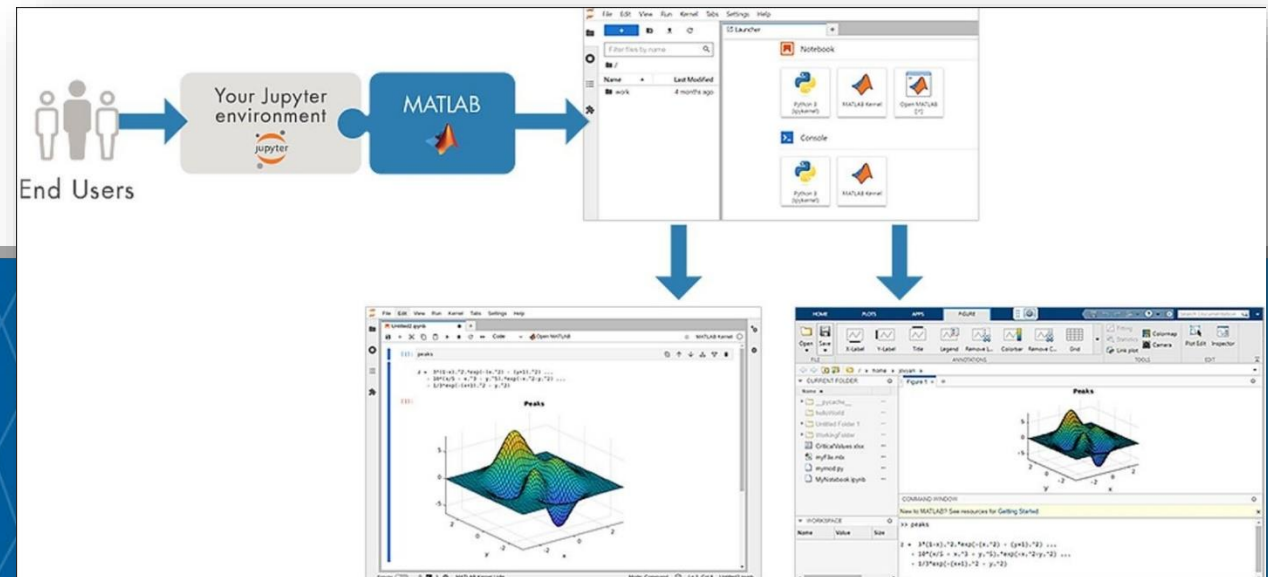


Integrating MATLAB with OOD: New Tools to Boost Usage and Improve User Experience

Nick Choi and Lisa Kempler, MathWorks
March 2025

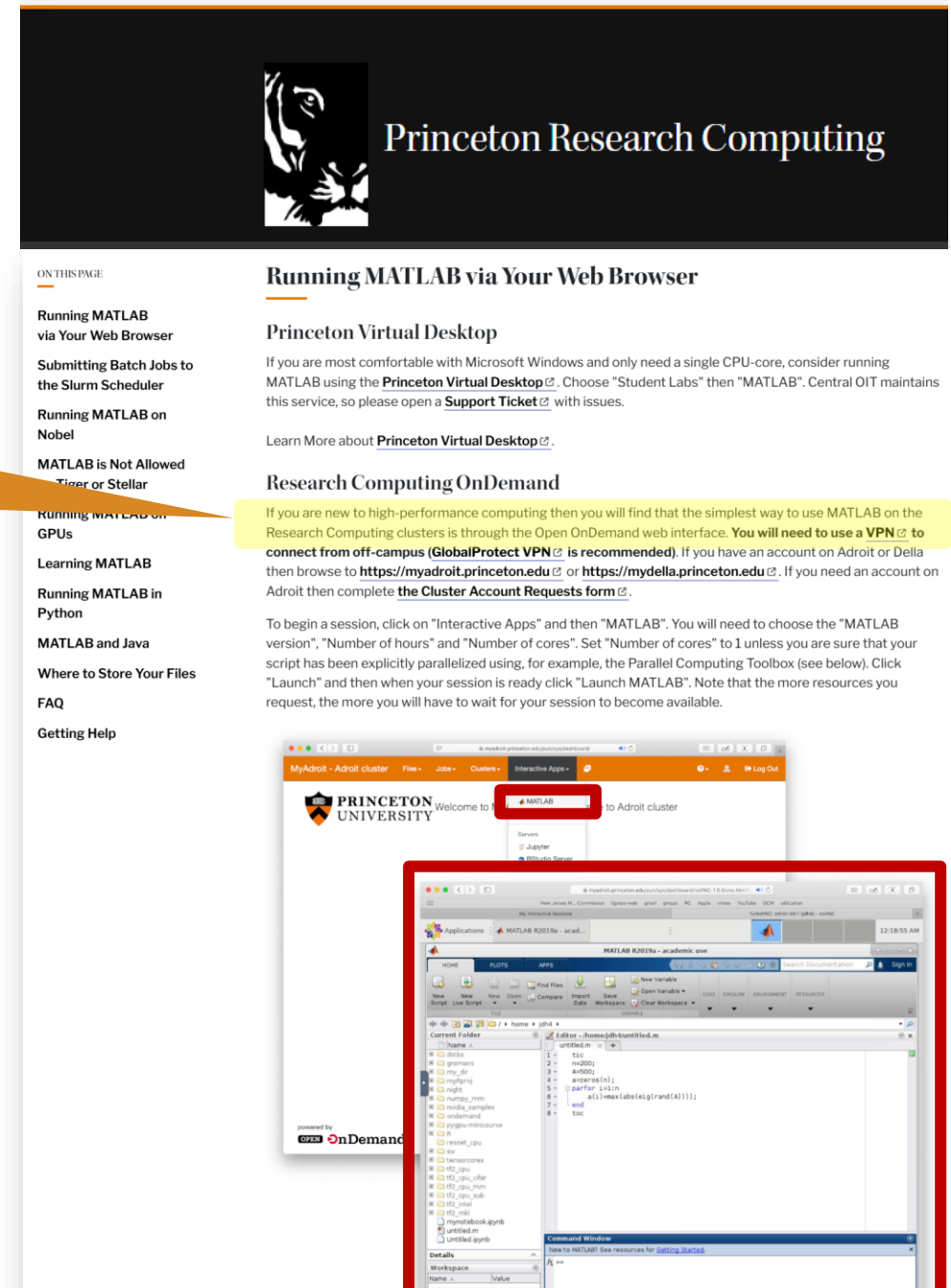


Open OnDemand + MATLAB Example: Princeton's Research Computing

"If you're new to HPC, you'll find the **simplest way** to use **MATLAB on HPC** clusters is through the **Open OnDemand web interface**"
Princeton Research Computing

Corollary: Common site feedback

After integrating MATLAB with OOD, usage + user satisfaction increase.



Princeton Research Computing

Running MATLAB via Your Web Browser

Princeton Virtual Desktop

If you are most comfortable with Microsoft Windows and only need a single CPU-core, consider running MATLAB using the [Princeton Virtual Desktop](#). Choose "Student Labs" then "MATLAB". Central OIT maintains this service, so please open a [Support Ticket](#) with issues.

Learn More about [Princeton Virtual Desktop](#).

Research Computing OnDemand

If you are new to high-performance computing then you will find that the simplest way to use MATLAB on the Research Computing clusters is through the Open OnDemand web interface. **You will need to use a VPN to connect from off-campus (GlobalProtect VPN is recommended)**. If you have an account on Adroit or Della then browse to <https://myadroit.princeton.edu> or <https://mydella.princeton.edu>. If you need an account on Adroit then complete the [Cluster Account Requests form](#).

To begin a session, click on "Interactive Apps" and then "MATLAB". You will need to choose the "MATLAB version", "Number of hours" and "Number of cores". Set "Number of cores" to 1 unless you are sure that your script has been explicitly parallelized using, for example, the Parallel Computing Toolbox (see below). Click "Launch" and then when your session is ready click "Launch MATLAB". Note that the more resources you request, the more you will have to wait for your session to become available.

Navigation Menu:

- Running MATLAB via Your Web Browser
- Submitting Batch Jobs to the Slurm Scheduler
- Running MATLAB on Nobel
- MATLAB is Not Allowed Tiger or Stellar
- Running MATLAB on GPUs
- Learning MATLAB
- Running MATLAB in Python
- MATLAB and Java
- Where to Store Your Files
- FAQ
- Getting Help

MyAdroit - Adroit cluster | **Interactive Apps** | **Log Out**

PRINCETON UNIVERSITY Welcome to **MATLAB** to Adroit cluster

Servers: **Apply** **Apply**

Applications: **MATLAB R2019a - academic**

Current Folder: **Editor - /home/joh-kuntiedt.m**

Workspace: **Workspace**

Command Window: **Now to MATLAB! See resources for [getting started](#)**

Poll Break

MATLAB in Open OnDemand: @ Ohio Supercomputer Center

The image shows the Ohio Supercomputer Center OnDemand web interface. On the left, a sidebar menu lists various applications. The 'Interactive Apps' menu item is highlighted with a yellow box. Below it, the 'MATLAB (legacy)' application is highlighted with a red box. Further down, 'Jupyter' and 'MATLAB' are also highlighted with red boxes. Arrows point from these red boxes to the main interface. The main interface shows the 'Recently Used Apps' section with four tiles: Jupyter (Shared by Kenneth Koepcke), Jupyter (System Installed App), MATLAB (legacy) (System Installed App), and MATLAB (System Installed App). An arrow points from the MATLAB tile to the bottom right, where a screenshot of the MATLAB web interface is shown. This interface displays a 'Notebook' section with five tiles: Python 3 (ipykernel), MATLAB Kernel, Open MATLAB [?], OSC PyTorch 2.4.0, and Python 3.12 (Conda 2024.06) [python/3.12]. Below this is a 'Console' section with four tiles: Python 3 (ipykernel), MATLAB Kernel, OSC PyTorch 2.4.0, and Python 3.12 (Conda 2024.06) [python/3.12].

Interactive Apps ▾ My Interactive Sessions ▾ All Apps ▾ Help ▾

Desktops

- Ascend Desktop
- Cardinal Desktop
- Ascend NextGen Desktop
- Pitzer Desktop
- Lightweight Desktop

GUIs

- ANSYS Workbench
- Abaqus/CAE
- Blender
- COMSOL Multiphysics
- IQmol
- MATLAB (legacy)**
- ParaView
- PyMOL
- QGIS
- Schrodinger
- Stata
- VMD
- Visit

Servers

- Code Server
- Jupyter**
- Jupyter + Spark
- MATLAB**
- RStudio Server

OSC OnDemand Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾ My Interactive Sessions ▾ All Apps ▾ Help ▾

Ohio Supercomputer Center

OnDemand provides an integrated, single access point for all of your HPC resources.

Recently Used Apps

- Jupyter
Shared by Kenneth Koepcke (kkoepcke)
- Jupyter
System Installed App
- MATLAB (legacy)
System Installed App
- MATLAB
System Installed App

<https://ondemand.osc.edu/node/c0316.ten.osc.edu/50168/lab/workspaces/auto-K>

Kernel Tabs Settings Help

Launcher

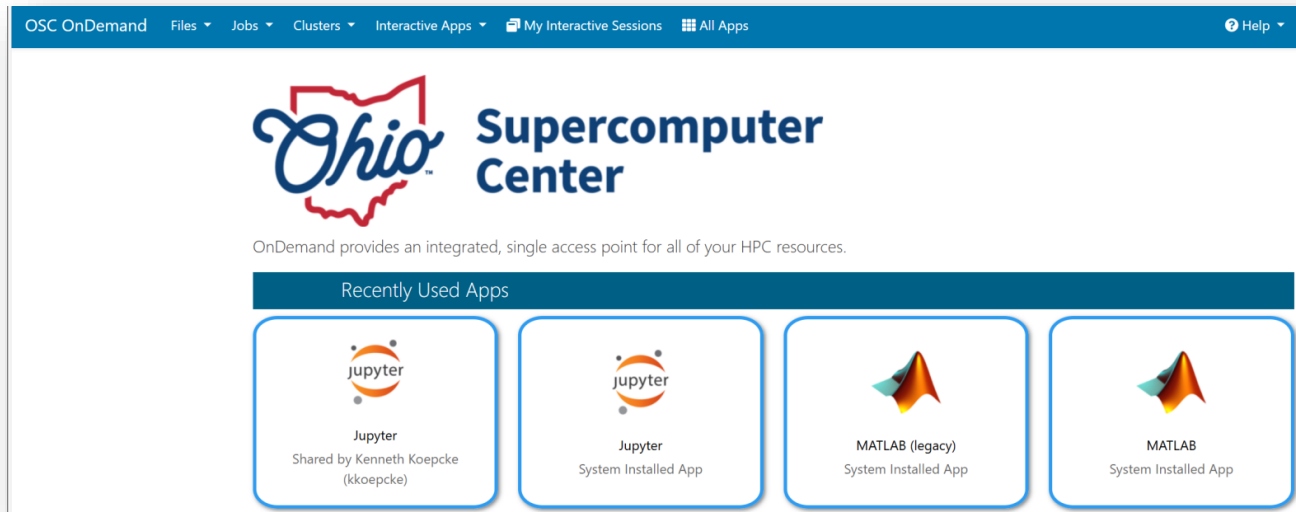
Notebook

- Python 3 (ipykernel)
- MATLAB Kernel
- Open MATLAB [?]
- OSC PyTorch 2.4.0
- Python 3.12 (Conda 2024.06) [python/3.12]

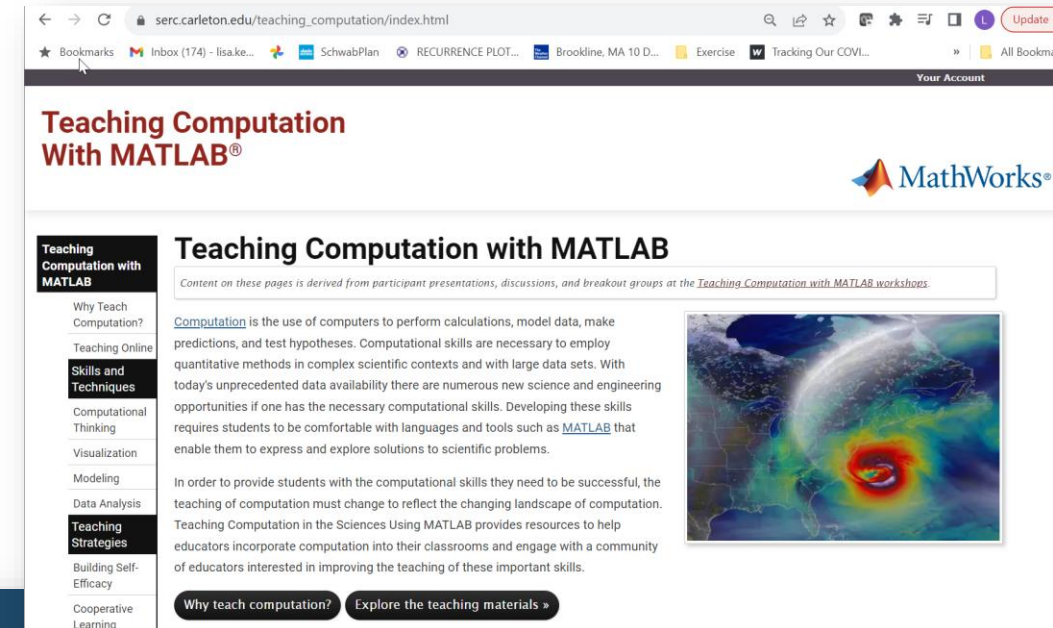
Console

- Python 3 (ipykernel)
- MATLAB Kernel
- OSC PyTorch 2.4.0
- Python 3.12 (Conda 2024.06) [python/3.12]

MATLAB Today: Integrations with Online Computing and Research Community Platforms



HPC and Research
Computing Centers



Resources for
Educators

Publishing papers
with MATLAB code



EU's ENES Data Space: MATLAB Desktop + via Jupyter Notebook

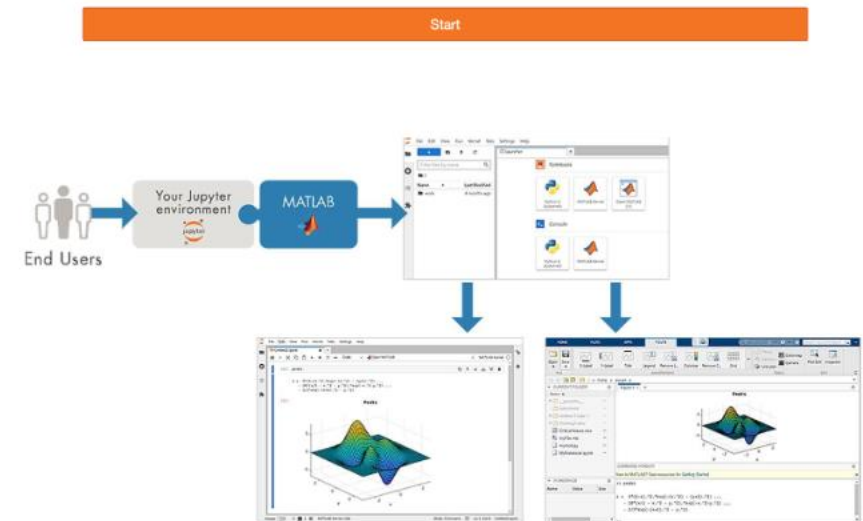


ENES Data Space

Home Notebooks Access Data Request

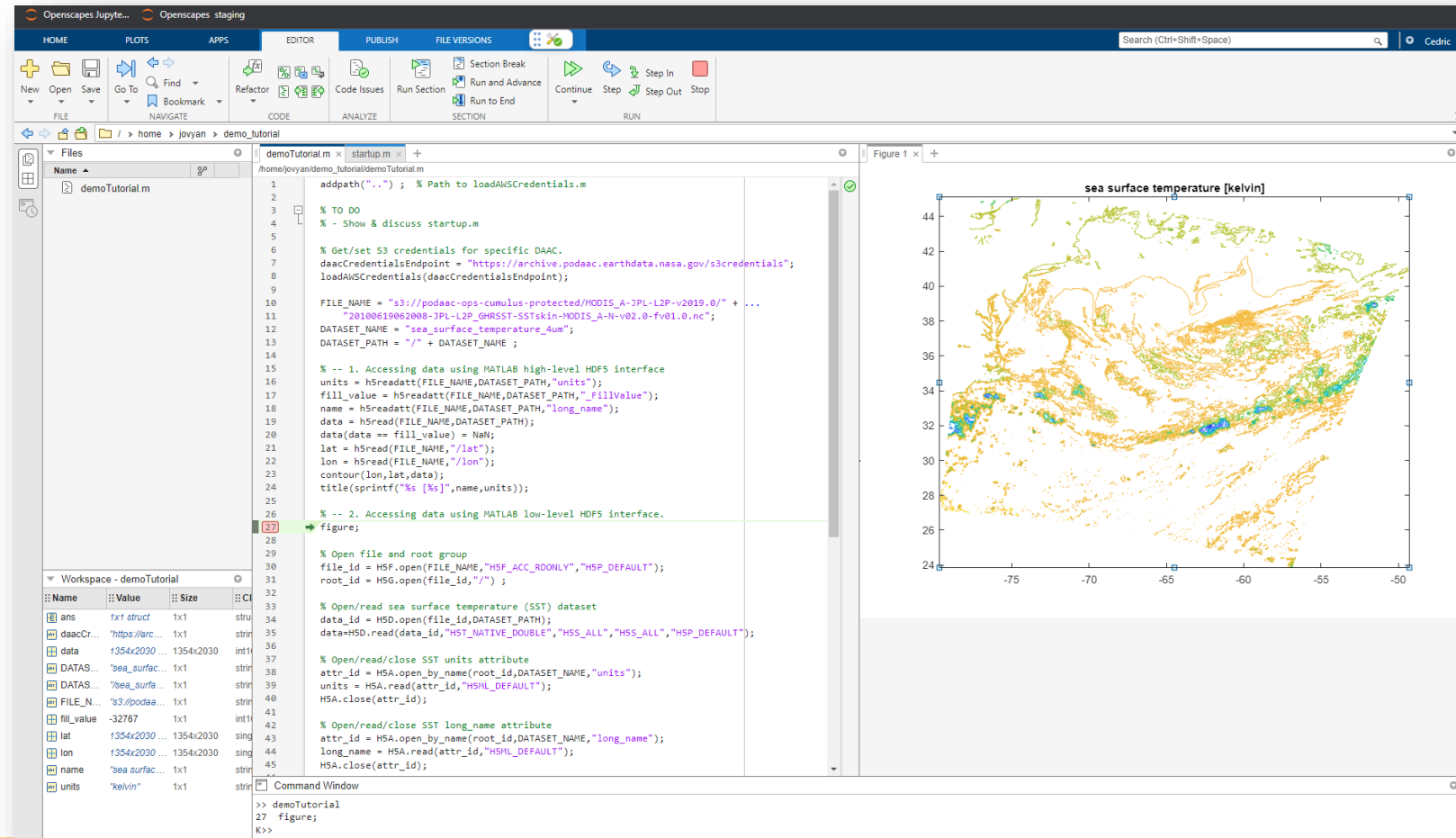
MATLAB Integration

Geo- and climate scientists around the globe [use MATLAB](#) to analyze, process and visualize climate data. Now, users of the ENES Data Space can login and start using MATLAB directly within the Jupyter-based environment, thus exploiting its powerful features to access and analyze data on the cloud. Users can either use the familiar MATLAB environment in a browser or the MATLAB kernel – available via the MATLAB Integration for Jupyter. An example to get started is available [here](#).



Use NASA Cloud Data with MATLAB on NASA Openscapes

2 Options: Desktop or Jupyter Notebook (JupyterHub 2i2c)



MATLAB on Openscapes

Accessing and Analyzing NASA Earthdata in the Cloud with MATLAB

MATLAB Live Script: Mapping Sea Surface Temperature

Blog: openscapes.org/blog/2023-10-17-matlab-on-openscapes/

Onramping Users: NASA Earthdata MATLAB Tutorial Example

Read Data

HDF5

[Python](#) [R](#) [MATLAB](#)

MATLAB provides low- and high-level programmatic interfaces for interacting with [HDF5](#) and [HDF4](#) files.

Focusing on HDF5, which is currently more prevalent, the high-level interface is made of seven high-level functions for getting information about HDF5 files and their content, creating datasets, and reading and writing datasets and attributes.

```
filename = "example.h5";  
h5disp(filename); % Display file content  
ds = "/g4/lat";  
lat = h5read(filename,ds);
```

The low-level interface is a thin wrapper around the underlying HDF5 C library. It is made of more than 300 functions organized in packages (H5F for file management, H5D for dataset management, etc.).

```
filename = "example.h5";  
ds = "/g4/lat";  
fID = H5F.open(filename);  
dsID = H5D.open(fID,ds);  
lat = H5D.read(dsID);  
H5D.close(dsID);  
H5F.close(fID);
```

On this page

- [netCDF](#)
- [HDF5](#)
- [GeoTIFF](#)
- [Shapefiles & GeoJSON](#)
- [CSV](#)
- [Excel](#)
- [MAT-Files \(.mat\)](#)
- [Text Data Files](#)
- [Binary Files](#)

[Edit this page](#)

NASA Earthdata Cloud Cookbook

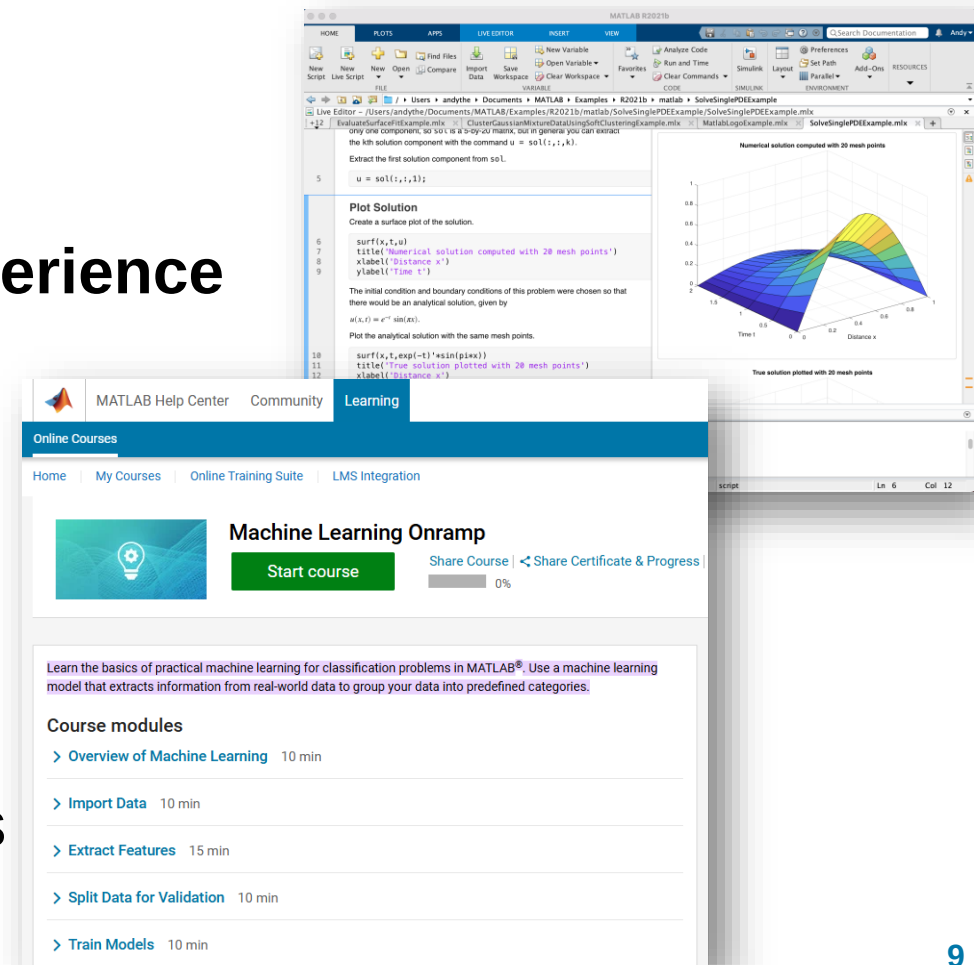
How do I...

- [find data](#)
- [access data in cloud](#)
- [access data locally](#)
- [read data](#)
- [subset data](#)
- [reformat data files](#)
- [reproject and regrid](#)
- [aggregate data](#)
- [plot data](#)
- [use APIs](#)
- [parking lot](#)

Cut and paste code
to reuse in MATLAB

Why MATLAB in OOD? User Appeal and Increased Usage

- >6500 universities and research institutes worldwide rely on MATLAB
 - Majority of those have MATLAB site licenses
 - Most have HPC resources
 - **OOD gives admins tools for better user experience**
 - Users require less HPC knowledge to ramp up
 - No SLURM, setup scripts
 - Eliminating those requirements will increase user adoption
 - HPC → HTC*
- *HTC = High Throughput Computing (Faster, but not necessarily parallel)
- Combine learning resources with OOD access
 - Ex: AI/ML/DL tutorials, MATLAB Onramps

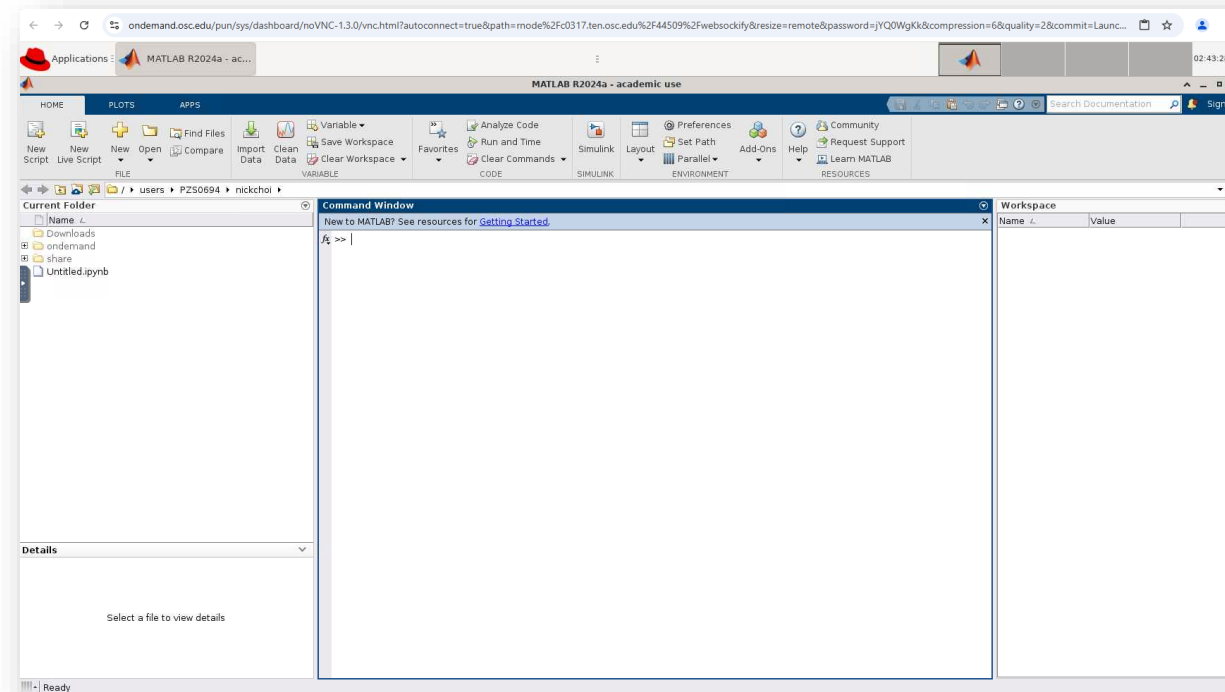


Agenda

- Why MATLAB in OOD?
- **Options for integrating MATLAB with OOD**
 - Best practices and other tips for integrating MATLAB with OOD
 - Helpful MATLAB features for MATLAB+OOD users
- Resources and Q&A
 - Documentation
 - Example implementations
 - MathWorks support resources

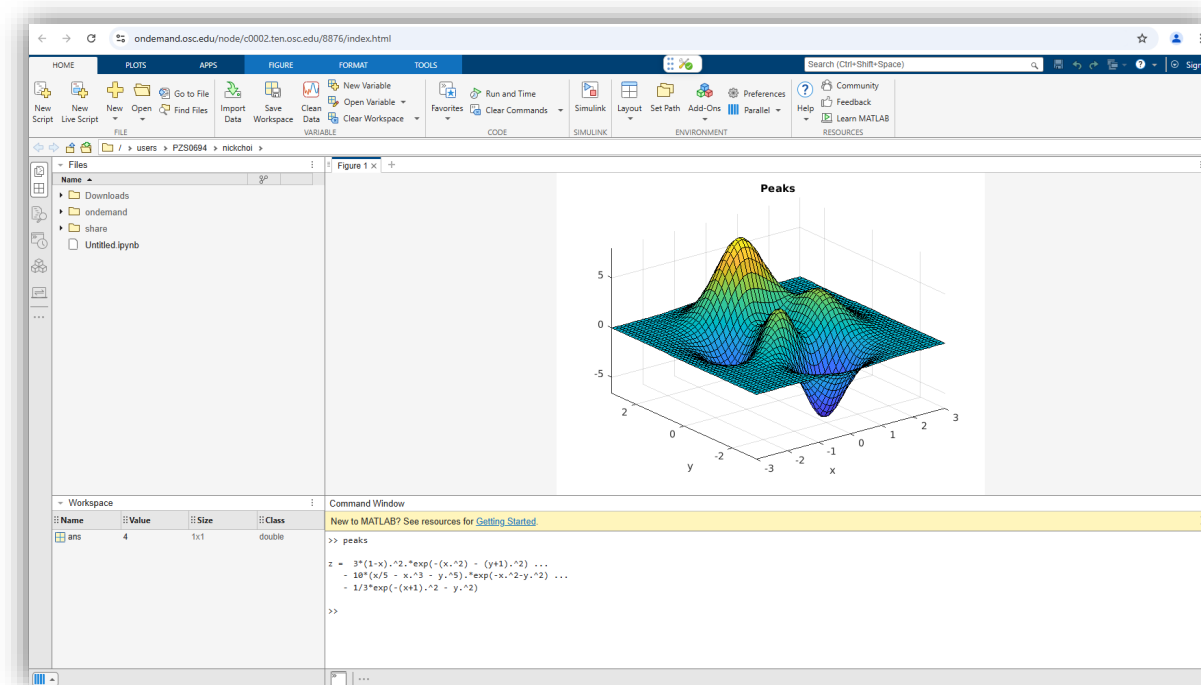
Multiple Approaches to Hosting MATLAB in Open OnDemand

Use Case	Use MATLAB
Non-interactive	With the <code>-batch</code> option
Interactive	As a command-line tool
	As a Linux desktop application (using noVNC or similar)



Multiple Approaches to Hosting MATLAB in Open OnDemand

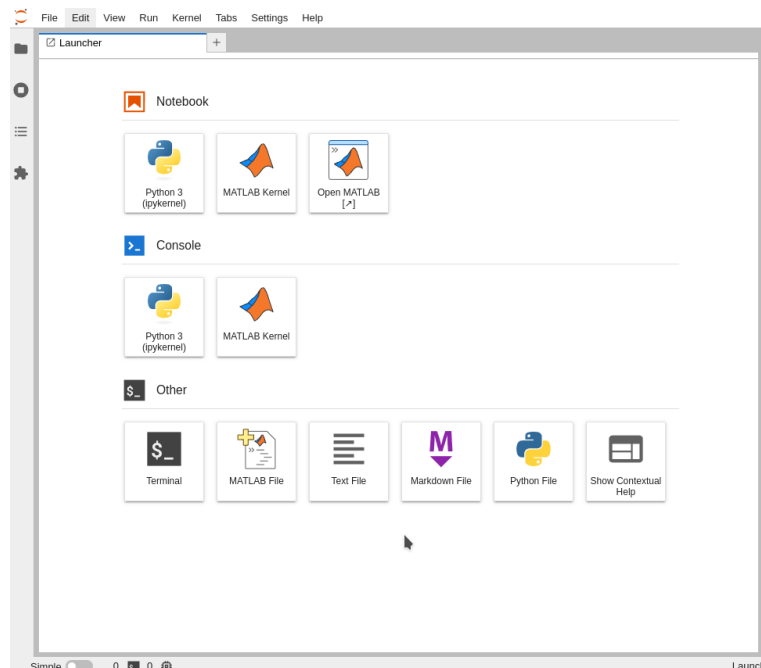
Use Case	Use MATLAB
Non-interactive	With the <code>-batch</code> option
Interactive	As a command-line tool
	As a Linux desktop application (using noVNC or similar)
	As a browser-based application (using matlab-proxy)



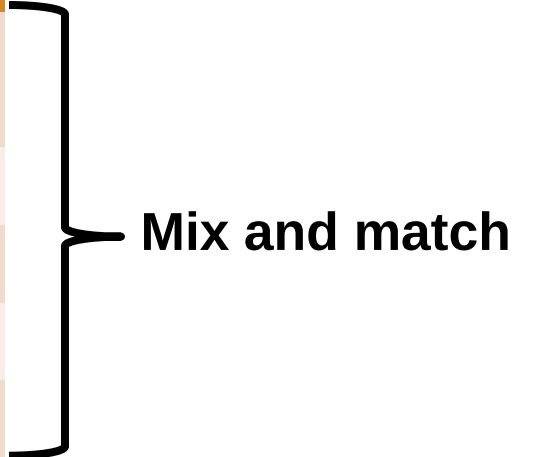
Multiple Approaches to Hosting MATLAB in Open OnDemand

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	As a browser-based application (using matlab-proxy)
	Through Jupyter

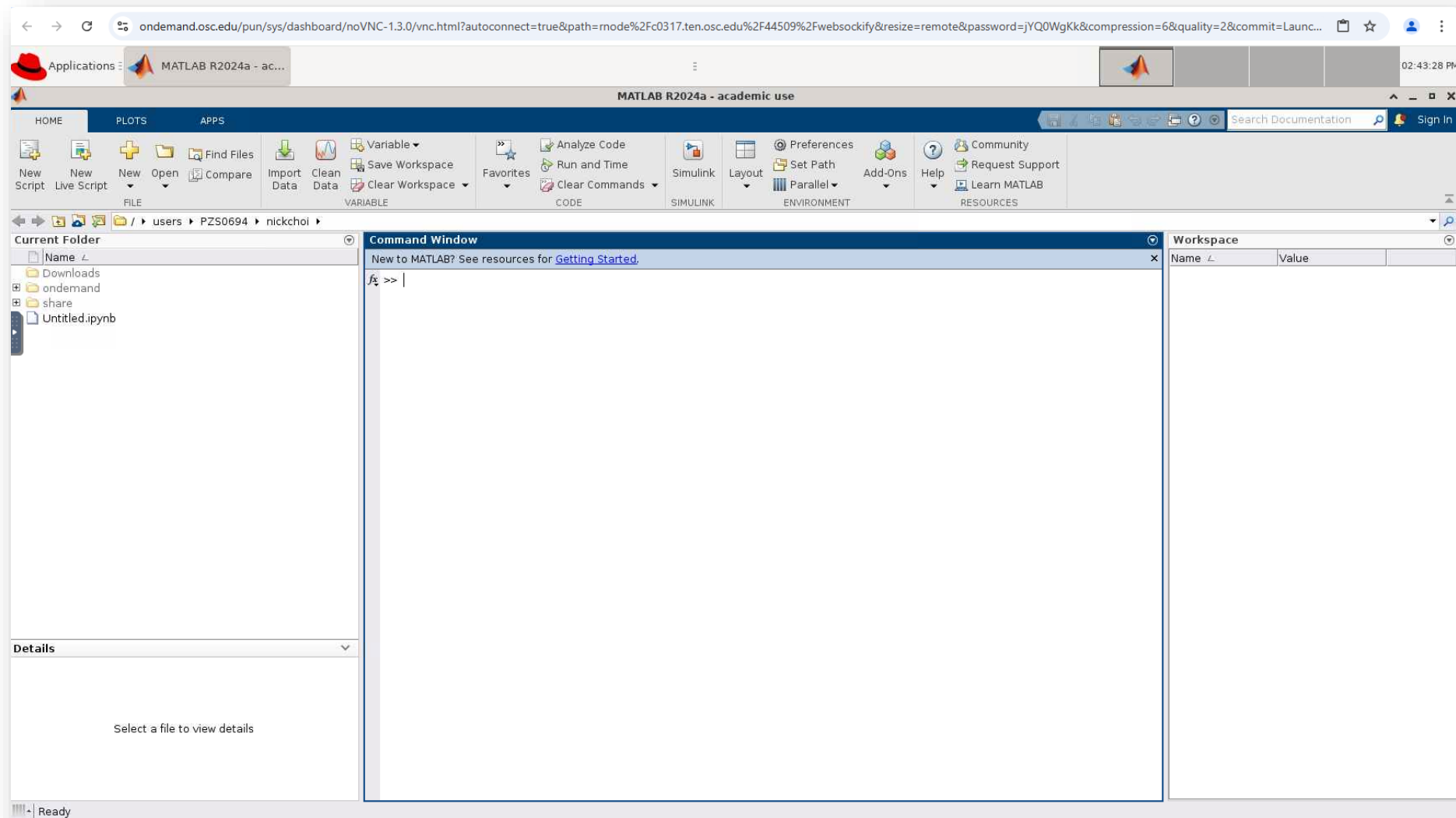
Mix and match



Multiple Approaches to Hosting MATLAB in Open OnDemand

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Non-interactive	With the <code>-batch</code> option	
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	Through Jupyter	

Accessing MATLAB Using a VNC-based approach

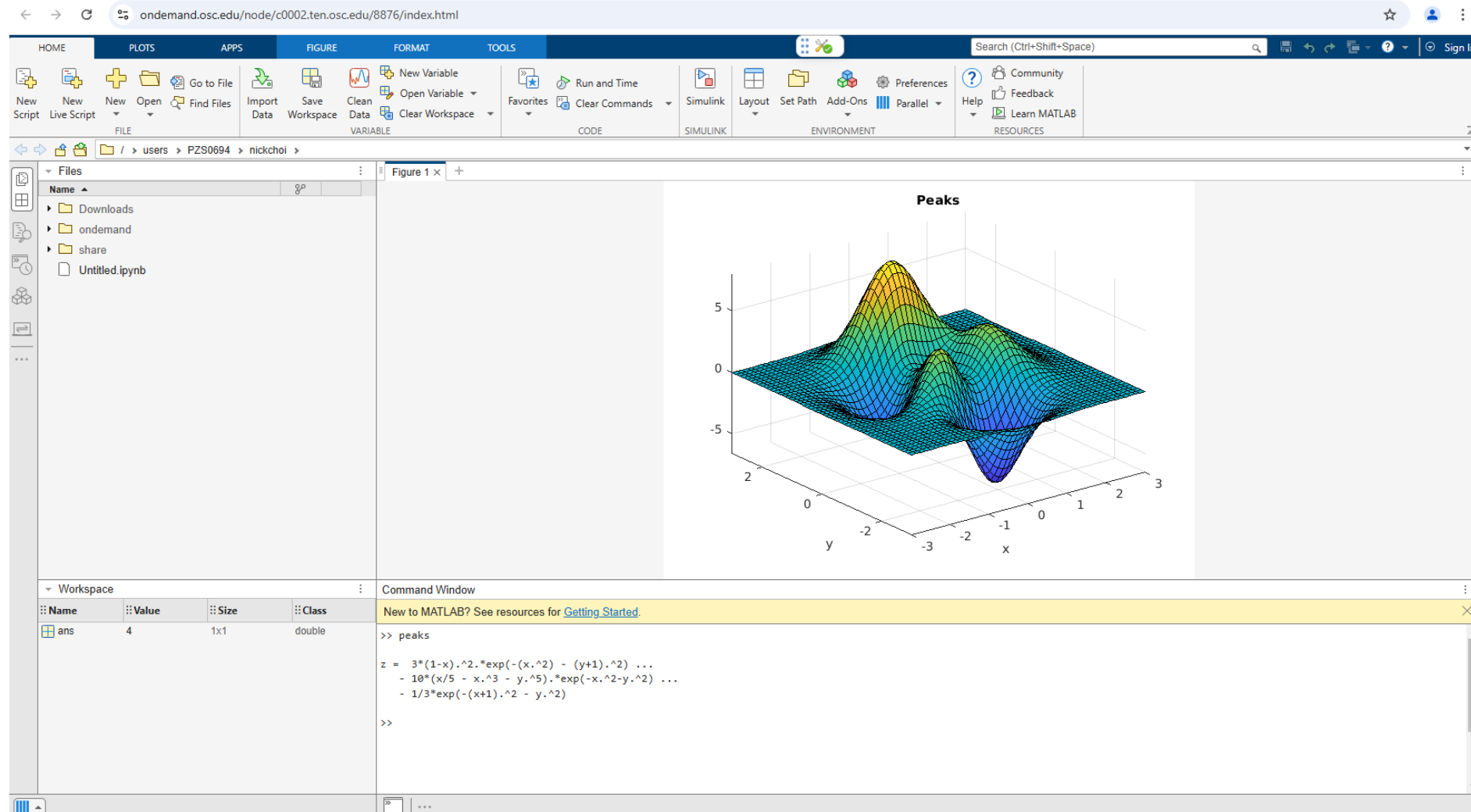


“The main issue for our users with running apps through VNC is that they are running in a Linux desktop context, so the keybindings are different, and copy/paste is not as easy in the browser...”

MATLAB as browser-based app with the matlab-proxy package

"This is the way forward for our usage of MATLAB – cleaner, faster, more direct"

"I definitely want to second though that the matlab-proxy is a huge improvement over having to run it in a virtual desktop"



Demo: Using MATLAB in Open OnDemand

The screenshot displays the Open OnDemand web interface in a browser. The address bar shows the URL `ondemand.osc.edu/pun/sys/dashboard/batch_connect/sessions`. The top navigation bar includes links for **OSC OnDemand**, **Files**, **Jobs**, **Clusters**, **Interactive Apps**, **My Interactive Sessions**, and **All Apps**. On the right of the navigation bar, there are links for **Help**, **Logged in as nickchoi**, and **Log Out**.

A green notification banner at the top states: "Session was successfully created." Below this, a breadcrumb trail shows **Home / My Interactive Sessions**.

The left sidebar contains two main sections:

- Shared Apps**: Includes a link for **Interactive Apps** and a specific app **Jupyter (kkoepcke)**.
- Interactive Apps**: Categorized into **Desktops** and **GUIs**.
 - Desktops**: Ascend Desktop, Cardinal Desktop, Ascend NextGen Desktop, Pitzer Desktop, Lightweight Desktop.
 - GUIs**: ANSYS Workbench, Abaqus/CAE, Blender, COMSOL Multiphysics, IQmol, MATLAB (legacy), ParaView, PyMOL, QGIS.

The main content area features a card for a **MATLAB (756106)** session. The card has a green header with the session name and status: **1 node | 16 cores | Running**. Below the header, the following details are listed:

- Host**: `>_c0326.ten.osc.edu` (with a red **Delete** button next to it).
- Created at**: 2025-03-12 14:24:33 EDT
- Time Remaining**: 32 minutes
- Session ID**: 212bf412-d0b6-4bb9-8d92-24f3ead42f69
- Problems with this session?** Submit support ticket

At the bottom of the card is a blue button labeled **Connect to Matlab**.

Tips for using the matlab-proxy package in OOD

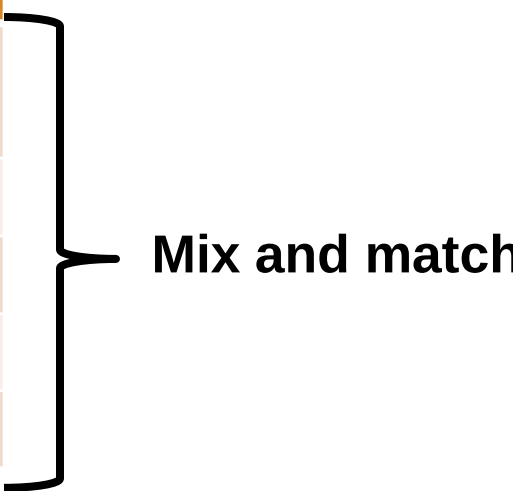
Many key behaviors can be configured using environment variables:

How can I...	Variable(s) to use
Specify where the app should run?	MWI_APP_PORT MWI_BASE_URL
Ensure everything is secure?	MWI_AUTH_TOKEN MWI_ENABLE_SSL_CERT_FILE MWI_SSL_KEY_FILE
Free up resources from idle sessions?	MWI_SHUTDOWN_ON_IDLE_TIMEOUT

Poll Break

Do you provide your users
access to GPUs through OOD?

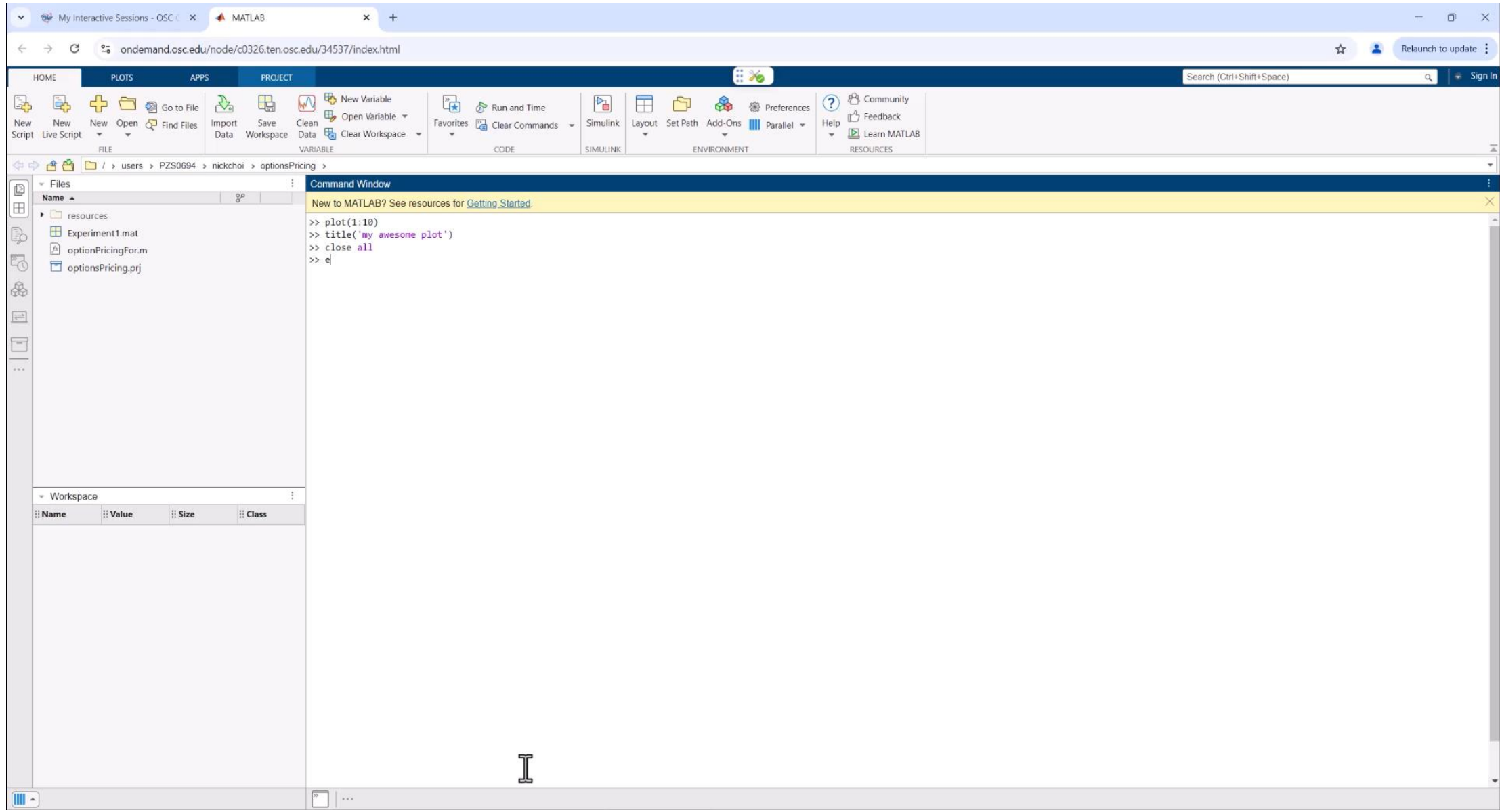
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	As a browser-based application (using matlab-proxy)	
	Through Jupyter	

Across all the approaches, MATLAB users can:

- ✓ Use multicore CPUs without needing to learn MPI
- ✓ Use GPUs without needing to learn CUDA
- ✓ Scale computations horizontally and vertically with MATLAB Parallel Server

Demo: Using MATLAB in Open OnDemand



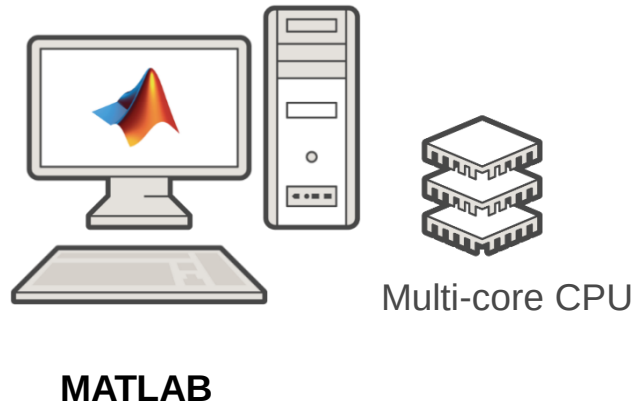
The screenshot displays the MATLAB Open OnDemand web interface. The browser's address bar shows the URL `ondemand.osc.edu/node/c0326.ten.osc.edu/34537/index.html`. The interface features a top navigation bar with tabs for **HOME**, **PLOTS**, **APPS**, and **PROJECT**. Below the navigation bar is a toolbar with various icons for file operations (New, Open, Save, etc.), code execution (Run, Clear), and environment management. The main workspace is divided into three panes:

- Left Pane (File Explorer):** Shows a tree view of the project structure. The current directory is `/ > users > PZS0694 > nickchoi > optionsPricing >`. The file list includes:
 - resources
 - Experiment1.mat
 - optionPricingFor.m
 - optionsPricing.prj
- Bottom-Left Pane (Workspace):** Displays a table of variables in the workspace. The table has columns for Name, Value, Size, and Class.

Name	Value	Size	Class
- Right Pane (Command Window):** Shows the MATLAB command prompt with the following code being executed:


```
>> plot(1:10)
>> title('my awesome plot')
>> close all
>> 
```

MATLAB has built-in multithreading



MathWorks®

MATLAB Multicore

Run MATLAB on multicore and multiprocessor machines

MATLAB® provides two main ways to take advantage of multicore and multiprocessor computers. By using the full computational power of your machine, you can run your MATLAB applications faster and more efficiently.

Built-in Multithreading

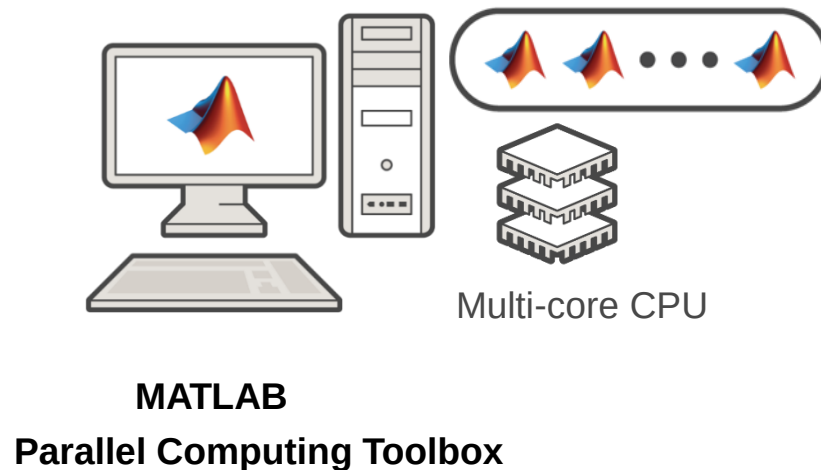
Linear algebra and numerical functions such as `fft`, `\(mldivide)`, `eig`, `svd`, and `sort` are multithreaded in MATLAB. Multithreaded computations have been on by default in MATLAB since Release 2008a. These functions automatically execute on multiple computational threads in a single MATLAB session, allowing them to execute faster on multicore-enabled machines. Additionally, many functions in Image Processing Toolbox™ are multithreaded.

Parallelism Using MATLAB Workers

You can run multiple MATLAB workers (MATLAB computational engines) on a single machine to execute applications in parallel, with [Parallel Computing Toolbox™](#). This approach allows you more control over the parallelism than with built-in multithreading, and is often used for coarser grained problems such as running parameter sweeps in parallel.

[MATLAB multicore](#)

Scale further with parallel computing



MATLAB Multicore

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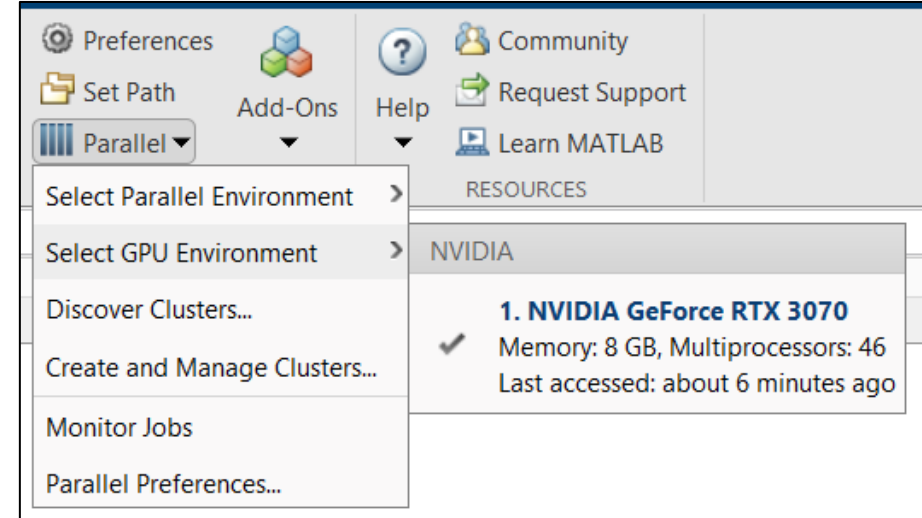
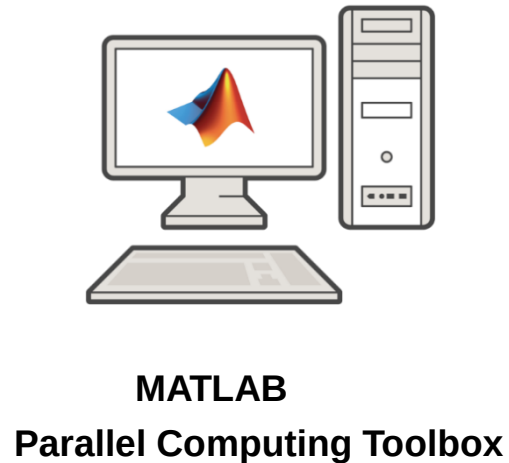
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Parallelism Using MATLAB Workers

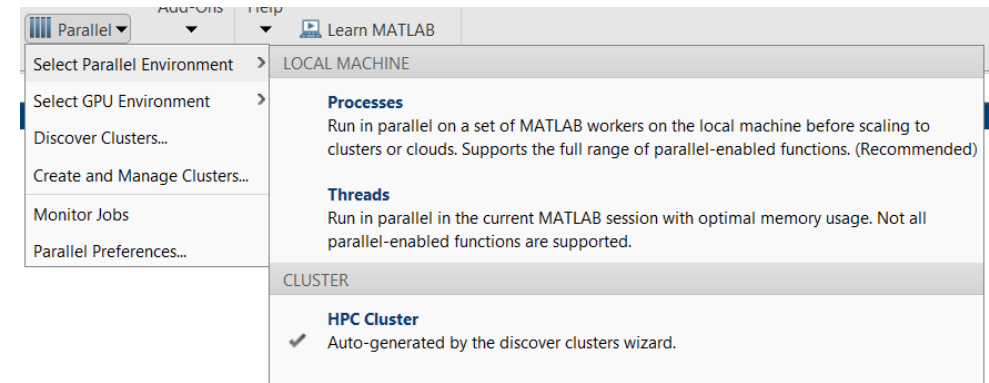
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Leverage NVIDIA GPUs without learning CUDA



With MATLAB Parallel Server you can scale to clusters and clouds

- Use more hardware with minimal code change
- Submit to on-premise or cloud clusters



Agenda

- Background
- Options for integrating MATLAB with OOD
 - Best practices and other tips for integrating MATLAB with OOD
 - Helpful MATLAB features for MATLAB+OOD users
- **Resources and Q&A**
 - Documentation
 - Example implementations
 - MathWorks support resources

Getting Started Resources

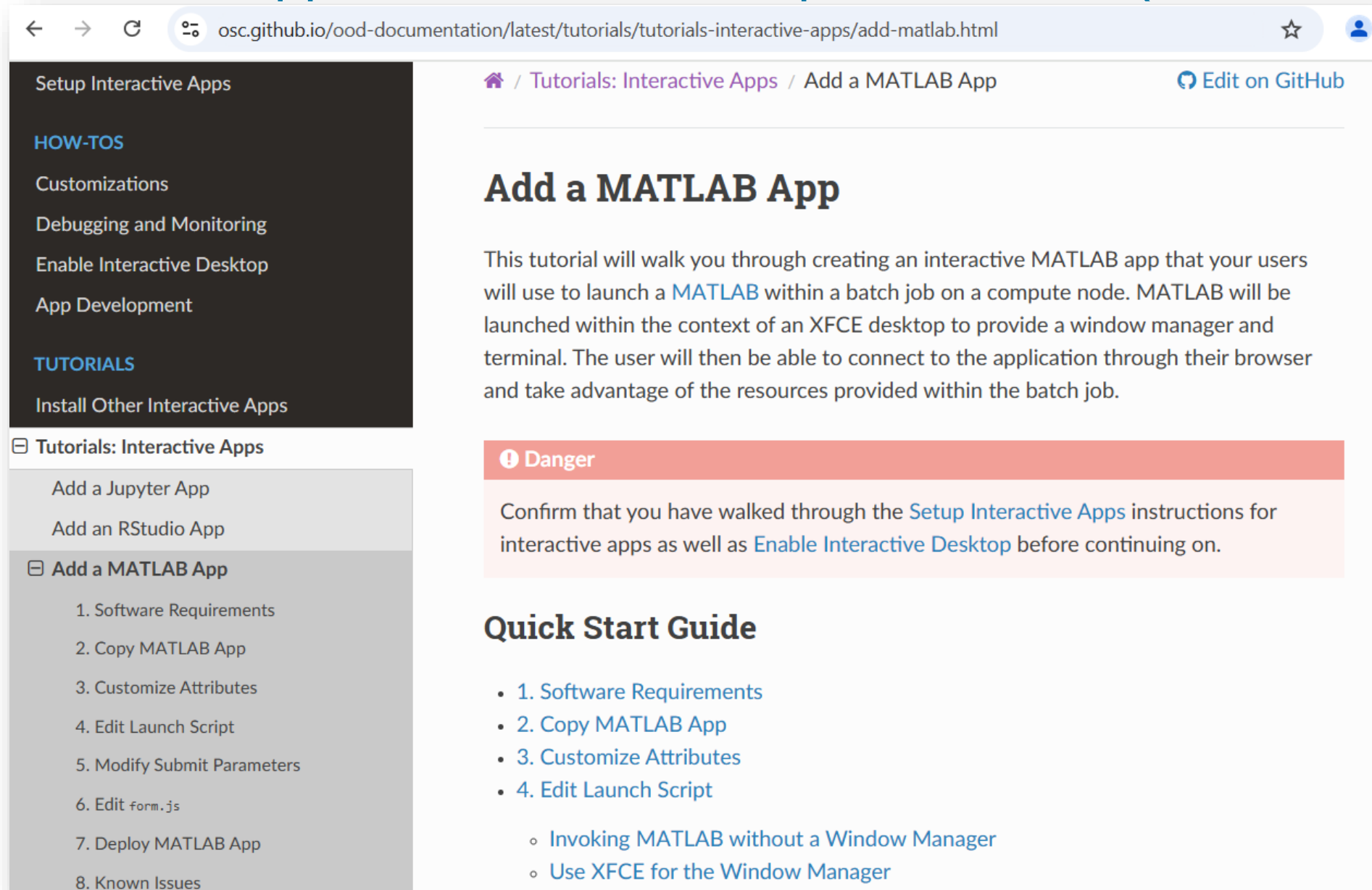
Visit these links to learn more about the topics discussed today:

- [Using the matlab-proxy package](#)
- [Using MATLAB Integration for Jupyter](#)
- [Using MATLAB for parallel computing](#)
- [Using MATLAB with GPUs](#)

To see how other sites used the matlab-proxy package in their MATLAB app:

- OSC: https://github.com/OSC/bc_osc_matlab_server
- UIUC – NCSA: <https://github.com/ncsa/Delta-OOD-MATLAB>

MATLAB OOD App Documentation Repo on GitHub (for admins)



The screenshot shows a web browser displaying the GitHub page for adding a MATLAB app. The browser's address bar shows the URL: `osc.github.io/ood-documentation/latest/tutorials/tutorials-interactive-apps/add-matlab.html`. The page has a dark sidebar on the left with navigation links. The main content area has a breadcrumb trail: `/ Tutorials: Interactive Apps / Add a MATLAB App`, followed by an `Edit on GitHub` link. The title is `Add a MATLAB App`. Below the title is a paragraph explaining the tutorial's purpose. A red warning box with a 'Danger' icon contains a confirmation message. At the bottom, there is a 'Quick Start Guide' section with a list of steps.

Setup Interactive Apps

HOW-TOS

- Customizations
- Debugging and Monitoring
- Enable Interactive Desktop
- App Development

TUTORIALS

- Install Other Interactive Apps

Tutorials: Interactive Apps

- Add a Jupyter App
- Add an RStudio App
- Add a MATLAB App**
 - 1. Software Requirements
 - 2. Copy MATLAB App
 - 3. Customize Attributes
 - 4. Edit Launch Script
 - 5. Modify Submit Parameters
 - 6. Edit `form.js`
 - 7. Deploy MATLAB App
 - 8. Known Issues

`/ Tutorials: Interactive Apps / Add a MATLAB App` [Edit on GitHub](#)

Add a MATLAB App

This tutorial will walk you through creating an interactive MATLAB app that your users will use to launch a [MATLAB](#) within a batch job on a compute node. MATLAB will be launched within the context of an XFCE desktop to provide a window manager and terminal. The user will then be able to connect to the application through their browser and take advantage of the resources provided within the batch job.

Danger

Confirm that you have walked through the [Setup Interactive Apps](#) instructions for interactive apps as well as [Enable Interactive Desktop](#) before continuing on.

Quick Start Guide

- 1. [Software Requirements](#)
- 2. [Copy MATLAB App](#)
- 3. [Customize Attributes](#)
- 4. [Edit Launch Script](#)
 - Invoking MATLAB without a Window Manager
 - Use XFCE for the Window Manager

What's Next?

- Contribute your best practices to the OOD user guide
 - Share your MATLAB OOD app repo and guidance on Discourse
 - Community-sustained effort

- Are you interested in an online MATLAB + OOD workshop?
 - Topic: A deeper dive on integrating matlab-proxy with OOD
 - Timeframe: TBD

Questions?



Thank you!